

HUJI Research Monitor

Healthcare · Weekly Digest · Week 29 · July 13, 2026

EXECUTIVE SUMMARY

This week's digest highlights groundbreaking advancements across medical devices, precision therapeutics, and AI-driven diagnostics. Featured innovations include a first-in-human device for safer aortic valve replacements and allele-specific gene therapies for rare genetic disorders. We also showcase AI platforms accelerating drug discovery and novel diagnostic approaches for conditions ranging from fibromyalgia and infant food allergies to critical care imaging and personalized IBD management, demonstrating significant market opportunities for investors and industry partners.

#1

First-in-Human Mechanical Leaflet Modification of Native Aortic Stenosis During TAVR.

36/50

Main Researcher: Anna Turyan

[Medical Device] [Clinical]

"Novel device makes TAVR safer, expanding life-saving treatment for aortic stenosis patients."

WHY NOW

As TAVR adoption grows, mitigating risks like coronary obstruction is crucial, opening a significant market for safety-enhancing tools to broaden patient eligibility.

COMMERCIAL ANGLE

The expansion of mechanical leaflet modification tools from bioprosthetic valves to native aortic stenosis represents a significant new market segment for transcatheter valve intervention technologies.

<https://pubmed.ncbi.nlm.nih.gov/42340176/>

#2

Allele-specific antisense oligonucleotide treatment rescues

34/50

Main Researcher: Tamar Harel

[Drug Discovery] [Genomics] [Clinical]

"Precision gene therapy targets specific mutations, offering breakthrough treatments for rare genetic diseases."

WHY NOW

This allele-specific approach overcomes limitations of current gene therapies, unlocking high-value therapeutic pathways for previously untreatable conditions.

COMMERCIAL ANGLE

The ability to precision-target single-nucleotide mutations opens a high-value therapeutic pathway for rare genetic disorders and personalized genomic medicine.

<https://pubmed.ncbi.nlm.nih.gov/42239176/>

#3

Progress in structure prediction and design of adaptive immune receptors.

30/50

Main Researcher: Dina Schneidman-Duhovny

[Drug Discovery] [Immunology] [Software/AI] [Synthetic Biology]

"AI platform revolutionizes immunotherapy design, accelerating personalized cancer treatments."

WHY NOW

Advances in AI and synthetic biology are converging, making in silico design of highly specific immune receptors a near-term reality to address unmet cancer needs.

COMMERCIAL ANGLE

Developing AI-driven platforms for the precision design of synthetic antibodies and T-cell receptors for personalized cancer immunotherapies.

<https://pubmed.ncbi.nlm.nih.gov/42430862/>

#4

Strategic Template Filtering Accelerates Fragment-Based Peptide Docking.

26/50

Main Researcher: Ora Schueler-Furman

[Drug Discovery] [Software/AI] [Proteomics]

"AI-powered software dramatically speeds up peptide drug discovery, cutting costs and time."

WHY NOW

High-efficiency computational tools are critical as pharmaceutical companies rapidly adopt AI/ML to accelerate their drug discovery pipelines.

COMMERCIAL ANGLE

Licensing or integrating this high-efficiency docking algorithm into AI-driven drug discovery platforms to accelerate the design of peptide-based therapeutics.

<https://pubmed.ncbi.nlm.nih.gov/42299509/>

#5

Nitric oxide inhibition ameliorates cortical proteomic changes in the Cntnap2^{-/-} and Shank3^{Δ4-22} mouse models of autism spectrum disorder.

25/50

Main Researcher: Haitham Amal

[Drug Discovery] [Neuroscience] [Proteomics] [Genomics]

"New drug target discovered for autism: paving the way for first-in-class small molecule therapies."

WHY NOW

Identifying specific molecular pathways for genetically-linked autism offers a validated drug target for a condition with significant unmet medical need.

COMMERCIAL ANGLE

The identification of specific nitric oxide signaling pathways provides a validated pharmacological target for developing first-in-class small molecule treatments for genetically-linked autism spectrum disorders.

<https://europepmc.org/article/MED/42010670>

#6

Haptoglobin, hemolysis, and renal vulnerability after pulsed field ablation: The hemoglobin-binding reserve.

23/50

Main Researcher: Rabea Asleh

[Medical Device] [Diagnostics] [Clinical]

"Diagnostic tool reduces kidney risk during cardiac ablation, enhancing patient safety for a growing procedure."

WHY NOW

Pulsed Field Ablation is a rapidly emerging cardiac therapy; a companion diagnostic that prevents serious complications will be in high demand.

COMMERCIAL ANGLE

Development of a pre-operative diagnostic screening tool or companion diagnostic to identify patients with low haptoglobin reserves who require protective measures before PFA.

<https://pubmed.ncbi.nlm.nih.gov/42235779/>

#7

Serum from Fibromyalgia Patients Activates Satellite Glial Cells in Mouse Peripheral Ganglia.

23/50

Main Researcher: Suhail Amar

[Diagnostics] [Drug Discovery] [Neuroscience] [Immunology]

"Blood-based biomarker panel for fibromyalgia promises objective diagnosis and targeted therapies."

WHY NOW

Fibromyalgia lacks objective diagnostics and effective treatments; identifying circulating disease factors represents a major step towards personalized medicine for millions.

COMMERCIAL ANGLE

Development of a blood-based diagnostic biomarker panel or targeted immunotherapies to neutralize the specific serum factors causing glial activation.

<https://pubmed.ncbi.nlm.nih.gov/42274567/>

#8

Strain-level microbial signatures and inferred functional alterations in infants with food protein-induced allergic proctocolitis.

23/50

Main Researcher: Moran Yassour

[Diagnostics] [Genomics] [Immunology] [Software/AI]

"Microbiome diagnostics for infants can prevent lifelong food allergies and guide early interventions."

WHY NOW

Growing understanding of the gut microbiome's role in immune development creates a timely opportunity for precision diagnostics and preventative therapeutics for infant allergies.

COMMERCIAL ANGLE

The identified strain-level microbial signatures provide a foundation for developing precision diagnostic kits and microbiome-based therapeutic interventions to prevent lifelong food allergies.

<https://pubmed.ncbi.nlm.nih.gov/42010622/>

#9

Temporal patterns of smartphone-mediated digital engagement and mental health symptomatology.

23/50

Main Researcher: Amit Birenboim

[Software/AI] [Diagnostics] [Neuroscience] [Clinical]

"Smartphone AI predicts mental health crises in real-time, enabling proactive digital interventions."

WHY NOW

Ubiquitous smartphone use and advanced AI capabilities create a powerful, scalable platform for continuous mental health monitoring and personalized support.

COMMERCIAL ANGLE

Developing digital phenotyping software for real-time mental health monitoring and predictive intervention via smartphone sensor data.

<https://pubmed.ncbi.nlm.nih.gov/42364396/>

#10

Multi-omics-based machine learning model predicts response and guides treatment in Crohn's disease: a case study in nutritional therapy-response to editorial.

22/50

Main Researcher: Dan Turner

[Diagnostics] [Software/AI] [Genomics] [Clinical]

"AI-driven precision nutrition model personalizes Crohn's disease treatment for better patient outcomes."

WHY NOW

The shift towards personalized medicine combined with multi-omics data offers a unique opportunity to optimize IBD management and improve patient quality of life.

COMMERCIAL ANGLE

Development of a companion diagnostic software platform for precision nutrition and personalized medicine in Inflammatory Bowel Disease (IBD) management.

<https://europepmc.org/article/MED/42133586>

#11

Dual-energy and photon-counting computed tomography in critical care.

22/50

Main Researcher: Sharon Einav

[Medical Device] [Diagnostics] [Imaging] [Clinical]

"Next-generation CT imaging offers superior diagnostics in critical care, improving patient outcomes."

WHY NOW

The demand for enhanced diagnostic accuracy and patient safety drives the adoption of advanced imaging technologies like photon-counting CT in critical care settings.

COMMERCIAL ANGLE

The transition toward photon-counting CT hardware and specialized reconstruction software presents a high-value market for next-generation diagnostic imaging systems in critical care.

<https://pubmed.ncbi.nlm.nih.gov/41962419/>

#12

Surgery for Pain Symptoms in Women with Isolated Superficial Peritoneal Endometriosis.

21/50

Main Researcher: Uri P Dior

[Diagnostics] [Imaging] [Clinical]

"Non-invasive diagnostic for endometriosis could prevent unnecessary surgeries and improve patient care."

WHY NOW

Endometriosis often leads to delayed diagnosis and invasive procedures; a precise, non-invasive screening tool addresses a significant unmet need for millions of women.

COMMERCIAL ANGLE

There is a significant commercial opportunity for improved non-invasive diagnostic tools or imaging modalities to accurately identify superficial endometriosis before surgery, reducing unnecessary procedures.

<https://pubmed.ncbi.nlm.nih.gov/41935570/>